AegisDrive-V v1.0.0

Integrated safety, security, MBSE and V-model engineering case study

Educational research; not a safe-vehicle certification claim

System and trust boundary

C++ ADAS, Brake and Body ECUs; Python longitudinal SiL

Camera/radar fusion and shared physical actuator

64-byte CAN FD HMAC envelope; public test keys, trusted local clock/FFI

Independent review scope

Same automation in separate reviewer roles, not accredited external assessment

Frozen v0.4 source/evidence; 92 baseline tests

846 non-host-timing outputs reproduced with zero mismatches

Eighteen security collision timelines

Strict: 6 stale-delivery +4 counter-gap +2 rollback/replay

Recovery: the same 6 stale-delivery failures

All matched A cases avoid collision; withholding plus rejection/timeout removes braking

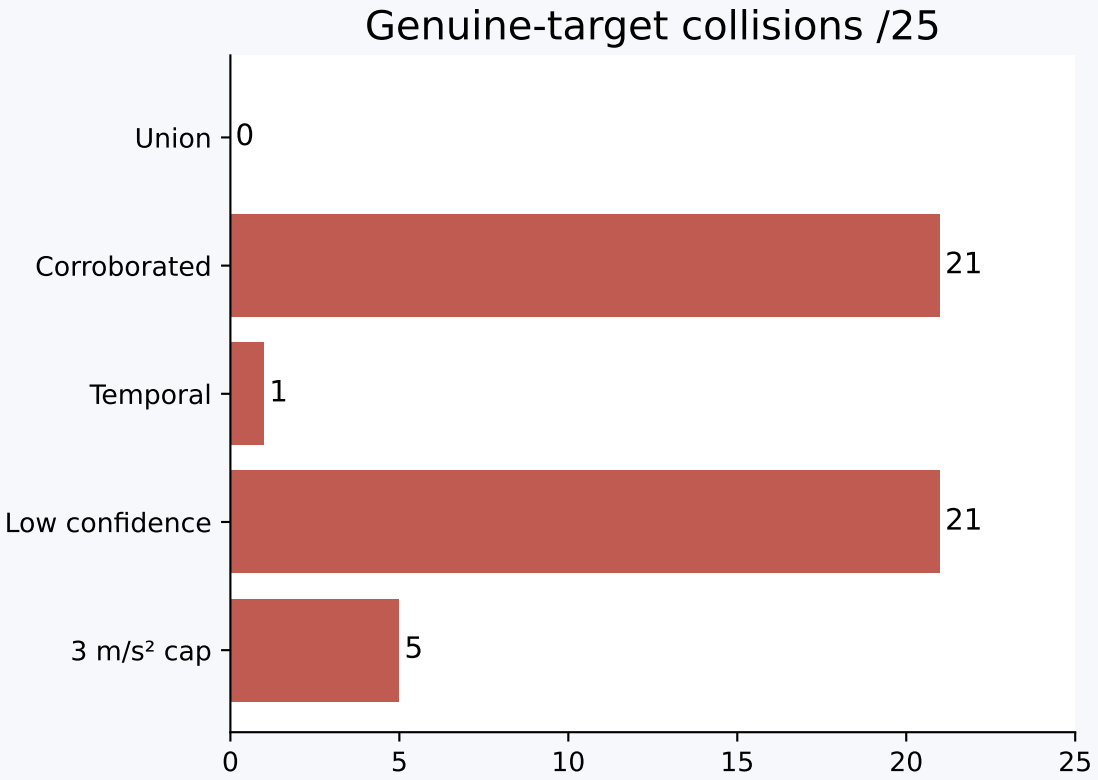
Scoped engineering changes

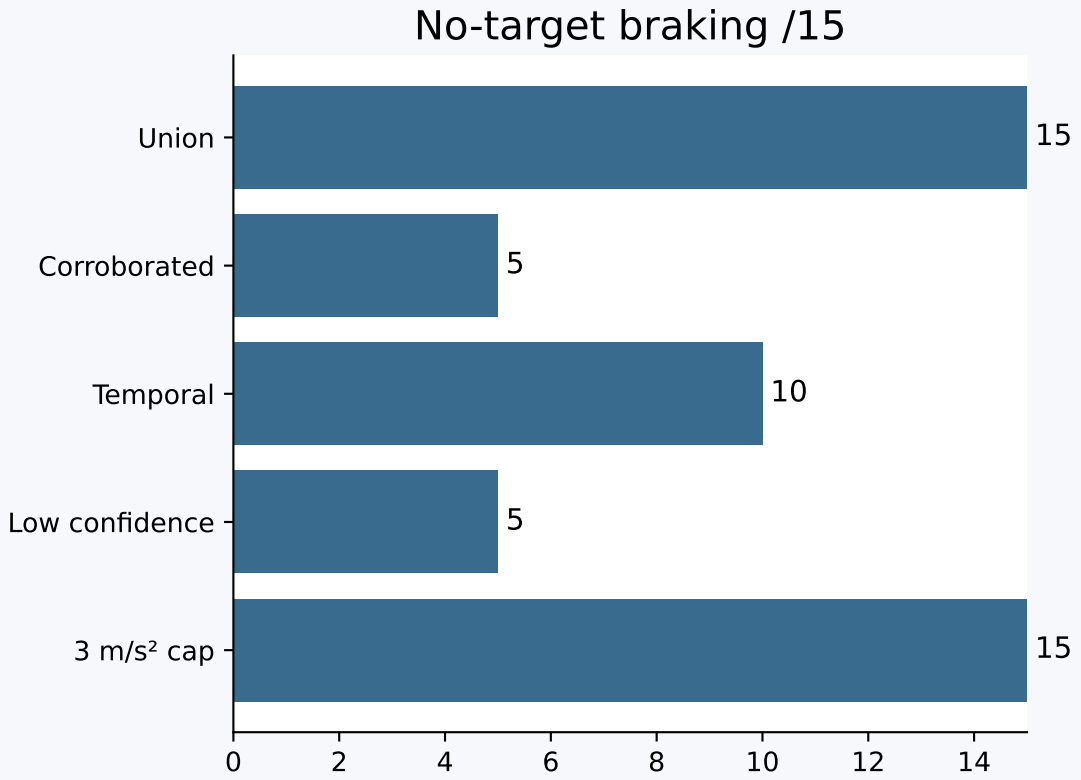
Select newer authenticated producer time across valid transports

No raw/stale/replayed command bypass

Current-pressure observation and 100 ms clear-dwell recovery; native DTC stays latched

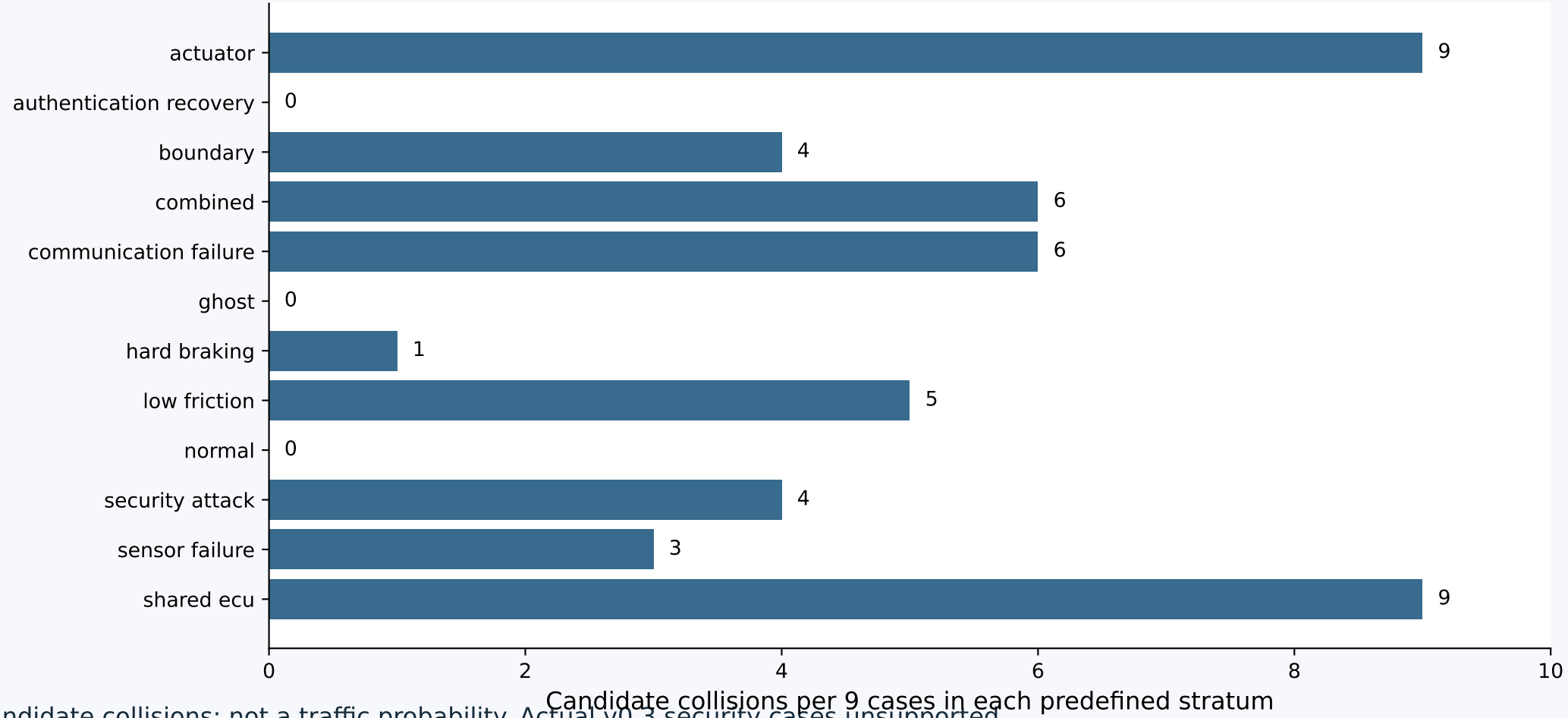
Perception tradeoffs





Designed cases; exact denominators differ. No following-vehicle or real traffic risk model.

Final robustness evidence



47/108 candidate collisions; not a traffic probability. Actual v0.3 security cases unsupported.

Verification and model limits

Requirement test status is separate from vehicle objective closure

Official SysML parse/semantic validation is not behavioral equivalence

Static analyzer and UBSan evidence; ASan runtime timeout, partial MC/DC retained

Structured assurance case

Claims + evidence + assumptions + counterexamples

Legacy HARA/goal/requirement derivation gaps remain explicit

Passing tests do not close unsafe vehicle objectives

Open issues and contribution

Shared ECU/actuator and security dependencies; trusted compromise

Ghosts, grip, ODD, short distance; no real follower/target WCET

Contribution: defensible decisions, reproducibility, traceable evidence and transparent risks